MAZHARUL ISLAM

Product Engineer

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SUMMARY

I build backend services, web and mobile applications, and applied AI workflows. I have 3.5 years of production experience at LII Lab and am now studying Computer Science at Paderborn University. Based in Paderborn, Germany, I am open to working-student roles in software development, backend engineering or applied AI, up to 20 hours per week.

TECHNICAL SKILLS

Languages and Frameworks: Python · Django · Django REST Framework · FastAPI · Dart · Flutter · TypeScript · Next.js · React

Databases and Messaging: PostgreSQL · Redis · Celery · Celery Beat

AI and NLP: Speech Recognition · NLP Processing · LLM Integration · PyTorch · WhisperX · OpenAI API

Infrastructure and DevOps: Docker · Nginx · GCP · DigitalOcean · CI/CD · Prometheus · Grafana · Firebase · FCM

AI-Assisted Development: Claude Code · Cursor · Agentic Workflows · AI Pair Programming

PROFESSIONAL EXPERIENCE

LII Lab

Sylhet, Bangladesh

Senior Software Engineer 01/2025 - 04/2026

Led backend engineering for OneIELTS and OnePTE, AI-powered English test preparation platforms.

- Architected a multi-tenant SaaS backend with django-multitenant: full data isolation, custom domain routing, per-tenant scoring config and feature flags.
- Designed a QTI 3.0-compliant exam engine supporting 100+ question types across IELTS Academic and General, with XML parsing, interaction routing, and strategy-pattern band-score normalization.
- Built a real-time speech evaluation pipeline integrating speech recognition, pronunciation assessment and NLP processing, with end-to-end scoring in under 15 seconds.
- Engineered multi-gateway payment infrastructure across 5 providers, with subscription lifecycle management and webhook idempotency.
- Built the Studio API layer for the moderator platform: content authoring, expert evaluation queue, exam rejudge pipelines, and KPI analytics.
- Set up production observability with Prometheus and Grafana: health monitoring, per-API latency instrumentation, and alerting.
- Mentored junior engineers through code reviews, PR feedback, and architectural walkthroughs.

Stack: Django · DRF · Python · PostgreSQL · Redis · Celery · FastAPI · Docker · GCP · Prometheus · Grafana · Next.js · TypeScript

LII Lab

Sylhet, Bangladesh

Software Engineer 11/2022 - 12/2024

Full-stack ownership of OnePTE: designed and shipped the Flutter app from day one and built the Django AI scoring backend.

- Designed and built the Flutter cross-platform app from scratch: all four PTE modules with 20+ task types, audio recording, and timer management.
- Built a task-group-based mock test engine with modular exam templates, configurable time allocations, automated question progression, and multi-dimensional score breakdowns.
- Engineered the AI scoring backend for spoken and written tasks using speech recognition, pronunciation assessment and NLP processing for multi-trait evaluation.
- Integrated subscription billing across 4 platforms with webhook handling, transaction deduplication, and in-app purchase verification.
- Built the Django admin and private API for content moderation, question bank management, subscription analytics with regional reporting, and user acquisition dashboards.
- Contributed to 93+ production releases across Android, iOS and Web during the full November 2022 to April 2026 LII Lab tenure; managed release pipelines, Firebase config, and CI/CD across all channels.

Stack: Flutter · Dart · Riverpod · Django · DRF · Python · PostgreSQL · Redis · Celery · Firebase · Next.js · TypeScript

PROJECTS

Sushi Lab, Bilingual restaurant ordering and marketing platform, Chartres, France 2026

- Ordering engine with a hydration-safe cart, scheduled-ahead time slots, per-item option pickers, and server-enforced availability windows
- Triple-channel notifications for every order, reservation and contact submission, dispatched in parallel with strict cross-channel data parity
- AI-discoverable SEO: JSON-LD, dynamic sitemap, dynamic OG images, per-locale canonical and hreflang, and llms.txt
- GA4 ecommerce funnel plus reservation, contact, language-switch and Web Vitals events

Stack: Next.js 16 · React 19 · TypeScript · Tailwind · Zustand · React Hook Form · Zod · React Email · next-intl · Telegram Bot API · GA4

Projecto, University course and proposal management system 2022

- Three roles with distinct permissions: student, supervisor and admin
- PDF and CSV export for project reports
- Background job service for scheduled operations

Stack: Django · Python · PostgreSQL · AJAX · jQuery · Bootstrap

TestForge, Mutation-driven test-backfill agent 2026

- Runs mutmut, reads the surviving mutants, and writes targeted tests to kill them
- Coverage is a proxy; mutation score is the thing coverage was trying to measure

Stack: Python · Claude API · mutmut · pytest

COMPETITIVE PROGRAMMING AND ACHIEVEMENTS

2000+ problems solved · 300+ contests · 1603 cf rating · 1965 cc rating

Champion, Champion at the National Hackathon, LU CSE Carnival

Outstanding Leadership Volunteer Award, IEEE Outstanding Leadership Volunteer Award

Runner-up, Bangladesh, Runner-up in Bangladesh at IEEEXtreme 16.0

Champion, Champion at LU TechStorm 4

51st, Dhaka Regional Main Round, Ranked 51st at the ICPC Dhaka Regional 2020 Main Round

Expert, 1600+, Reached Expert on Codeforces

Honorable Mention, Honorable Mention at the ICPC Asia Dhaka Regional Online Preliminary

Codeforces (Max 1603) · CodeChef (Max 1965) · LightOJ · Toph · StopStalk · Vjudge

EDUCATION

Paderborn University Paderborn, Germany

M.Sc. Computer Science · Enrolled for Winter Semester 2026/27. Oct 2026 - present

Leading University Sylhet, Bangladesh

B.Sc. in Computer Science and Engineering · CGPA 3.60 / 4.00 Degree awarded 2023

VOLUNTEERING

IEEE Computer Society, LU Student Branch Chapter

Chair Apr 2022 - May 2023

- Co-founded the chapter and launched its first community initiatives
- Organized seminars, webinars and technical events with industry speakers
- Grew membership by more than 20% through outreach programmes
- Delivered a multi-session Flutter bootcamp
- Served as judge and problem setter for programming contests

Leading University Computer Club

ACM Coordinator Jan 2022 - Dec 2023

- Conducted peer-to-peer sessions on data structures and algorithms
- Mentored juniors in competitive programming across Codeforces, CodeChef and LightOJ
- Organized regular contest practice to prepare members for ICPC and national competitions

Leading University Sports Club

Joint General Secretary 2022 - 2023

- Elected to the executive committee
- Organized inter-department tournaments and university sporting events